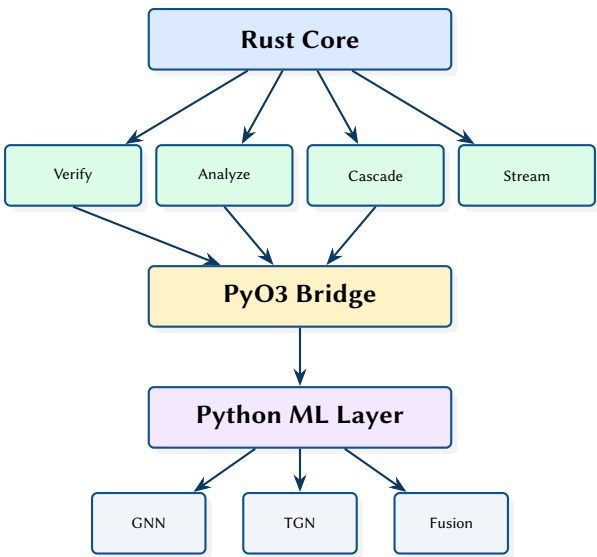


NetAssure: A GNN-Based Real-Time Network Analytics Platform

Formal Verification, Cascade Simulation, and Anomaly Detection

Simon Knight

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Abstract. This report presents NetAssure, a hybrid Rust/Python network analytics platform that provides six complementary analysis paradigms: formal verification via Header Space Analysis, graph-theoretic metrics, Monte Carlo failure cascade simulation, GNN-based anomaly detection, real-time streaming with WebSocket alerting, and topology optimisation. The Rust core (netassure-core, netassure-verify, netassure-analyze, netassure-cascade) handles performance-critical deterministic algorithms, while a Python layer provides PyTorch-based machine learning including temporal graph networks and multi-modal fusion scoring. PyO3 bindings bridge the two worlds, exposing verification, analysis, and cascade simulation to the Python ML pipeline.

Version	Date	Changes
0.1.0	March 2026	Initial architecture report

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List of Design Decisions & Rules

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1 Introduction and Motivation

Network operators need multiple analysis perspectives to understand complex topologies: formal verification alone cannot predict cascade failures, while machine learning alone cannot guarantee reachability invariants. NetAssure integrates six complementary paradigms into a single platform with a unified topology model.

The six paradigms are:

1. **Formal Verification** — Header Space Analysis ([HSA](#)), reachability proofs, loop detection, and traffic isolation checking.
2. **Graph Algorithms** — Centrality, community detection, path diversity, and robustness metrics.
3. **Failure Cascade Modelling** — Monte Carlo simulation with load redistribution and scenario comparison.
4. **Machine Learning (ML)/Graph Neural Network (GNN)** — Anomaly detection, temporal graph networks, and feature attribution via GNNExplainer.
5. **Real-Time Streaming** — Inference pipeline, anomaly detector, alert engine, and WebSocket streaming.
6. **Optimisation** — Critical node identification and redundancy recommendations.

: Hybrid Rust/Python Architecture

Deterministic algorithms (verification, graph metrics, cascade simulation) are implemented in Rust for performance and correctness. Machine learning workloads run in Python (PyTorch/PyTorch Geometric) where the ecosystem is mature. PyO3 bridges the two at the function call level, avoiding serialisation overhead.

2 System Architecture

2.1 Crate Structure

The Rust workspace contains four library crates and a CLI binary:

Table 1. NetAssure Rust crate organisation.

Crate	Role	Key Dependencies
netassure-core	Topology model, ingestion	petgraph, serde
netassure-verify	HSA , reachability	bit-set, petgraph
netassure-analyze	Graph algorithms	rustworkx-core
netassure-cascade	Monte Carlo simulation	rayon, rand
netassure-py	PyO3 bindings	pyo3
cli	Unified CLI	clap, axum, tokio

2.2 Topology Model

The core topology is a `petgraph::StableGraph` with typed node and edge attributes. Stable indices ensure that node/edge removal does not invalidate existing references—critical for cascade simulation where nodes are progressively disabled.

```
pub struct Topology {
```

```

graph: StableGraph<NodeAttr, EdgeAttr>,
node_index: HashMap<String, NodeIndex>,
}

pub struct NodeAttr {
pub id: String,
pub node_type: NodeType,
pub capacity: f64,
pub load: f64,
}

```

2.3 REST API

The ingestion system exposes an Axum-based HTTP Application Programming Interface (API) for real-time topology synchronisation and prediction queries:

Table 2. REST API endpoints.

Method	Path	Description
GET	/health	Liveness probe
GET	/health/ready	Readiness (topology synced?)
GET	/topology	Full topology JSON
GET	/predictions	GNN node predictions
GET	/anomalies	Recent anomaly history

3 Formal Verification

The verification engine implements HSA [1] over the topology graph. Each forwarding rule is modelled as a header-space transfer function; the engine computes reachability by composing transforms along all paths between source and destination.

Design Decision 1: Bit-Set Header Representation

Header spaces are represented as bit-sets rather than ternary vectors. This trades some expressiveness for $O(1)$ intersection and union operations, which dominate the verification runtime. For the network sizes targeted (hundreds of nodes, thousands of rules), this approach is sufficient and avoids the complexity of BDD-based representations.

Loop detection operates by tracking visited nodes during header-space propagation. If a packet’s header matches a forwarding rule that would send it to an already-visited node, a loop is reported with the full cycle path.

4 Failure Cascade Simulation

The cascade simulator models progressive network failure through load redistribution. When a node or link fails, its traffic is redistributed across remaining paths. If any node exceeds its capacity threshold, it fails in turn, potentially triggering a cascade.

```

pub fn simulate_cascade(
topo: &mut Topology,
initial_failures: &[NodeIndex],
redistribution: RedistributionPolicy,
capacity_threshold: f64,

```

```

) -> CascadeReport {
  let mut round = 0;
  let mut failed: HashSet<NodeIndex> = initial_failures.iter().cloned().collect();
  loop {
    let overloaded = find_overloaded(&topo, &failed, capacity_threshold);
    if overloaded.is_empty() { break; }
    failed.extend(overloaded);
    redistribute_load(topo, &failed, &redistribution);
    round += 1;
  }
  CascadeReport { rounds: round, total_failed: failed.len(), .. }
}

```

Monte Carlo mode runs thousands of iterations with random initial failure sets (parameterised by failure percentage), using Rayon for parallel execution. Each iteration produces a cascade depth and scope, enabling statistical analysis of network resilience.

5 Machine Learning Pipeline

5.1 GNN Anomaly Detection

The **GNN** model [2] operates on a 14-dimensional feature schema per node (degree centrality, betweenness, load ratio, interface counts, etc.) and produces per-node anomaly scores.

5.2 Temporal Graph Networks

The Temporal Graph Network (**TGN**) model processes time-series topology events (link flaps, load changes) to predict future anomalies. It maintains a per-node memory vector updated at each event, enabling the model to capture temporal patterns such as recurring failures or gradual degradation.

5.3 Multi-Modal Fusion

The fusion scorer combines three signal types:

- **Topology embeddings** (14-dim feature schema)
- **Metrics sequence** (5-dim: latency, loss, jitter, throughput, error rate)
- **Log embeddings** (768-dim from a pre-trained encoder)

Design Decision 2: Fusion Architecture

The fusion model concatenates the three embedding vectors after per-modality projection heads, then passes through a two-layer MLP with dropout. This late-fusion approach allows each modality to be independently pre-trained and evaluated, simplifying debugging when one signal source is noisy or unavailable.

***Insight:**
Scenario comparison mode runs two cascade experiments with different topologies (e.g. before and after adding redundancy) and produces a diff report highlighting improvements in cascade depth, scope, and critical node rankings.*

6 Design Summary

Table 3. Key design decisions and their rationale.

Decision	Rationale
Hybrid Rust/Python	Deterministic algorithms in Rust for speed; ML in Python for ecosystem maturity
PyO3 bindings	Zero-copy where possible; avoids JSON serialisation on the hot path
StableGraph	Node removal during cascade simulation must not invalidate existing indices
Bit-set HSA	$O(1)$ set operations; sufficient for target network sizes
Late fusion scoring	Independent modality training; graceful degradation when a signal is missing

References

- [1] P. Kazemian, G. Varghese, and N. McKeown, “Header space analysis: Static checking for networks,” *NSDI*, 2012.
- [2] T. N. Kipf and M. Welling, “Semi-supervised classification with graph convolutional networks,” in *ICLR*, 2017.